

Laparoscopic Surgery Repositories on GitHub

A Comprehensive Analysis: Video Datasets vs. Extracted Image Datasets

Introduction

This document catalogs GitHub repositories related to laparoscopic surgery, categorized into two groups: (1) repositories that contain or link to actual surgical video datasets, and (2) repositories that work with image frames extracted from surgical videos. For each repository, the dataset size, storage requirements, number of extractable examples, and software dependencies are listed.

Category A: Repositories with Video Datasets

These repositories contain or provide direct access to full laparoscopic surgery video datasets. Videos are typically hosted externally (due to GitHub file size limits) and accessed via download links.

1. CAMMA-public/TF-Cholec80

<https://github.com/CAMMA-public/TF-Cholec80>

Description:	80 cholecystectomy (gallbladder removal) surgery videos by 13 surgeons
Videos:	80 full-length laparoscopic surgery videos
Total Frames:	370,168 frames extracted at 1 FPS
Storage Required:	166 GB to download; 85.2 GB after extraction
Annotations:	Phase labels (7 phases) and tool presence (7 tools) per frame
Software Req.:	Python 3, TensorFlow 1.x or 2.x, tqdm, Matplotlib
Hardware Req.:	Ubuntu 20.04, CUDA 10.1, NVIDIA GTX1080Ti GPU recommended
Examples Extractable:	370,168 annotated frames from 80 videos

2. CAMMA-public/cholect45

<https://github.com/CAMMA-public/cholect45>

Description:	45 cholecystectomy videos with surgical action triplet annotations
Videos:	45 videos (subset of Cholec80)
Total Frames:	90,489 frames at 1 FPS
Annotations:	127,385 triplet instances (instrument-verb-target), tool, phase labels
Storage Required:	~50 GB estimated (subset of Cholec80 85.2 GB)
Software Req.:	Python 3, PyTorch or TensorFlow 2, Pillow
Examples Extractable:	90,489 frames with 127,385 triplet annotations

3. CAMMA-public/MultiBypass140

<https://github.com/CAMMA-public/MultiBypass140>

Description:	140 videos of laparoscopic Roux-en-Y gastric bypass surgery from 2 centers
Videos:	140 full surgical videos (70 from Strasbourg + 70 from Bern)
Annotations:	12 phases, 46 steps, 5 types of intraoperative adverse events
Storage Required:	~150-200 GB estimated for full video dataset
Software Req.:	Python 3, PyTorch, FFmpeg (for frame extraction)
Examples Extractable:	Frames at 1 FPS from 140 videos (tens of thousands of frames)

4. CAMMA-public/Endoscapes

<https://github.com/CAMMA-public/Endoscapes>

Description:	201 laparoscopic cholecystectomy videos for scene segmentation and CVS
Videos:	201 laparoscopic cholecystectomy videos
Total Frames:	58,813 frames (1 FPS) + 11,090 CVS-annotated frames

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Sub-datasets:	CVS201 (11,090 frames), BBox201 (1,933 frames), Seg50 (493 frames)
Storage Required:	~100+ GB estimated
Software Req.:	Python 3, PyTorch, OpenCV
Examples Extractable:	58,813 frames; 1,933 with bounding boxes; 493 with segmentation masks

5. CAMMA-public/cholectrack20

<https://github.com/CAMMA-public/cholectrack20>

Description:	20 surgical videos for multi-class multi-tool tracking
Videos:	20 full-length laparoscopic cholecystectomy videos (~14 hours total)
Total Frames:	35,000 frames at 1 FPS; 50,581 frames at 25 FPS
Annotations:	65,000+ tool instance labels with bounding boxes, IDs, phases
Storage Required:	~30-50 GB estimated
Software Req.:	Python 3, PyTorch, OpenCV
Examples Extractable:	35,000 annotated frames with 65,200 tool labels

6. franciszchen/VLSurg-dataset

<https://github.com/franciszchen/VLSurg-dataset>

Description:	Versatile Laparoscopic Surgical video dataset for general surgical AI
Videos:	60 patient surgery videos
Annotations:	Surgical phases, skill assessment scores, surgical descriptions
Storage Required:	~50-100 GB estimated (60 videos with frames)
Software Req.:	Python 3, PyTorch
Examples Extractable:	Thousands of frames from 60 patient videos with multi-task labels

7. CAMMA-public/ConvLSTM-Surgical-Tool-Tracker

<https://github.com/CAMMA-public/ConvLSTM-Surgical-Tool-Tracker>

Description:	Surgical tool tracking using ConvLSTM on Cholec80 videos
Videos:	Uses Cholec80 dataset (80 videos); links to YouTube demo videos
Storage Required:	85.2 GB (uses Cholec80 dataset)
Software Req.:	Python >=2.7, TensorFlow 1.3-2.0, FFmpeg, OpenCV, imageio, Matplotlib
Examples Extractable:	Tool tracking trajectories across all 80 Cholec80 videos

8. AI-Medical-Robotics/Surgical-Tool-Localization

<https://github.com/AI-Medical-Robotics/Surgical-Tool-Localization>

Description:	Surgical tool localization in endoscopic videos using deep learning
Videos:	24,000 video clips (109 GB) + m2cai16 (15 cholecystectomy videos)
Total Frames:	23,000 frames from m2cai16; 2,532 with spatial bounding boxes
Storage Required:	109 GB for Endoscopic Vision Challenge dataset
Software Req.:	Python, PyTorch, CUDA-capable GPU
Examples Extractable:	24,000 video clips + 23,000 labeled frames with 7 tool classes

9. ruaridhg/blender_laparoscopy

https://github.com/ruaridhg/blender_laparoscopy

Description:	Synthetic laparoscopic/colonoscopy data generated using Blender
Data Type:	Synthetic videos (.avi) and rendered image frames (.png)
Storage Required:	blender.zip ~61 MB, liver.zip ~6.09 GB, examples.blend ~15.67 MB
Total Storage:	~6.2 GB for all synthetic data

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Software Req.:	Blender 3.4.1, Python, Pandas, PyTorch (for UNet)
Hardware Req.:	Min: Dual-core CPU, 4GB RAM; Recommended: Quad-core, 16GB RAM, GPU
Examples Extractable:	Unlimited synthetic frames (rendered from Blender scenes)

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Category B: Repositories with Extracted Images from Videos

These repositories work with image frames that have been extracted from laparoscopic surgery videos. They do not provide access to the original full-length videos but use the extracted frames for machine learning tasks such as smoke detection, tool detection, and image classification.

1. nancy280/Smoke-Detection-in-Laprosopic-Surgery

<https://github.com/nancy280/Smoke-Detection-in-Laprosopic-Surgery>

Description:	Smoke detection in laparoscopic video frames using CNN + ML models
Source Videos:	10 robot-assisted laparoscopic hysterectomy procedure videos
Extracted Frames:	600 total (300 hazy images + 300 clear images)
Frame Rate:	1 FPS extraction from source videos
Storage Required:	<1 GB (small image dataset)
Software Req.:	Python, Jupyter Notebook, scikit-learn, OpenCV
ML Models Used:	CNN, Random Forest, XGBoost, SVM
Examples Extractable:	600 labeled images (binary: hazy vs. clear)

2. AakarshMishra/Smoke-Detection-in-Laprosopic-Surgery-main

<https://github.com/AakarshMishra/Smoke-Detection-in-Laprosopic-Surgery-main>

Description:	Fork of smoke detection project (same dataset as nancy280 repo)
Source Videos:	10 robot-assisted laparoscopic hysterectomy procedure videos
Extracted Frames:	600 total (300 hazy + 300 clear images)
Frame Rate:	1 FPS extraction
Storage Required:	<1 GB
Software Req.:	Python, Jupyter Notebook, scikit-learn, OpenCV
Examples Extractable:	600 labeled images

3. cchandel-dev/Laprosopic-Surgery-Work

<https://github.com/cchandel-dev/Laprosopic-Surgery-Work>

Description:	Multi-project repo: image annotation, tool detection, temporal ordering
Sub-projects:	(1) Private Image Annotator, (2) Surgical Tool Detection, (3) Temporal Ordering
Data Type:	Medical images (private, not publicly downloadable)
Storage Required:	Small (code + annotations only; images not included)
Software Req.:	Python, pip install -r requirements.txt
Examples Extractable:	Depends on user-supplied images; Siamese network for image ordering

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Summary: All Repositories at a Glance

#	Repository	Cat.	Videos	Frames/Examples	Storage	Type	Key Feature
1	CAMMA/TF-Cholec80	A	80 videos	370,168 frames	85.2 GB	Cholecystectomy	Phase + tool labels
2	CAMMA/cholect45	A	45 videos	90,489 frames	~50 GB	Cholecystectomy	Action triplets
3	CAMMA/MultiBypass140	A	140 videos	Tens of thousands	~150-200 GB	Gastric bypass	Multi-center
4	CAMMA/Endoscapes	A	201 videos	58,813 frames	~100+ GB	Cholecystectomy	Segmentation + CVS
5	CAMMA/cholectrack20	A	20 videos	35,000 frames	~30-50 GB	Cholecystectomy	Tool tracking
6	VL Surg-dataset	A	60 videos	Thousands	~50-100 GB	General laparo.	Multi-task AI
7	CAMMA/ConvLSTM-Tracker	A	80 (Cholec80)	370,168 frames	85.2 GB	Cholecystectomy	Tool trajectory
8	Surgical-Tool-Loc.	A	24K clips+15	23,000+ frames	109 GB	Cholecystectomy	Tool localization
9	blender_laparoscopy	A	Synthetic	Unlimited	~6.2 GB	Synthetic	Blender-generated
10	nancy280/Smoke-Det.	B	(from 10)	600 images	<1 GB	Hysterectomy	Smoke detection
11	AakarshM/Smoke-Det.	B	(from 10)	600 images	<1 GB	Hysterectomy	Smoke detection
12	cchandell/Lapro-Work	B	N/A	User-supplied	Small	Mixed	Annotation+detect.

Overall Summary & Conclusions

Total Repositories Found: 12

- Category A (Video Datasets): 9 repositories
- Category B (Extracted Images): 3 repositories

Total Storage Required (all datasets combined):

- Minimum estimate: ~520 GB
- Maximum estimate: ~700+ GB
- Largest single dataset: Cholec80 at 85.2 GB (166 GB download)
- Smallest dataset: Smoke Detection repos at <1 GB each

Total Examples Extractable:

- From video repos (Category A): ~950,000+ annotated frames across all datasets
- From image repos (Category B): ~1,200 labeled images
- Combined total: ~950,000+ usable training/testing examples

Key Observations:

1. The CAMMA-public organization (University of Strasbourg) is the dominant source of laparoscopic surgery datasets on GitHub, contributing 6 of the 12 repositories.
2. Most video datasets focus on laparoscopic cholecystectomy (gallbladder removal), which is the most commonly performed minimally invasive surgery worldwide.
3. GitHub repos typically do not host videos directly (GitHub has a 100 MB file limit). Instead, they provide code, annotations, and download links to external hosting.

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4. The largest publicly available dataset is Cholec80 (80 videos, 85.2 GB), while MultiBypass140 offers the most videos (140) from multiple surgical centers.
5. Common software requirements across all repos: Python 3, PyTorch or TensorFlow, OpenCV, and a CUDA-capable GPU for training deep learning models.
6. The Blender Laparoscopy repo is unique in offering synthetic data generation, allowing unlimited frame creation for training purposes.
7. The extracted-image repositories (Category B) are significantly smaller in scale but useful for specific tasks like smoke/haze detection in surgical visibility.